

On critical reflection.

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ABSTRACT

This paper presents an analysis and clarification of the major role of critical reflection of assumptions (CRA) in adult learning. It examines the differences among types of critical reflection, the role of CRA in the Transformation Theory of adult learning, philosophical foundations of CRA, its development over the lifespan, non-cognitive aspects, its function in both tacit and explicit decision-making, and its role in the validation of beliefs and expressions of feelings through discourse. The paper includes a taxonomy of applications of CRA and discussion of educational implications of this construct.

This article presents a concept analysis of critical reflection and particularly of critical reflection of assumptions (CRA). This concept is central to understanding how adults learn to think for themselves rather than act on the concepts, values, and feelings of others. It is also central to understanding many of the theories and practices common in the professional discourse of adult educators, including critical theory, critical thinking, dialectical thinking, critical literacy, critical pedagogy, the learning organization, action science, psychotherapy, and the Transformation Theory of adult learning (Mezirow, 1991, 1995, 1996). CRA is also central to the work of adult educators (Brookfield, 1987, 1995; Cranton, 1994; Mezirow, 1990; Watkins & Marsick, 1993). The literature of our field, however, has not fully recognized the crucial and generic role of this concept in adult learning.

This paper analyzes and clarifies the meaning, significance, development, and common applications of CRA. The intention is to help educators more fully understand the nature and importance of CRA to significant adult learning, the different ways CRA may be used programmatically, and its implications for learning theory and adult education.

Critical Reflection of Assumptions: Concept Differentiation

Reflection, a "turning back" on experience, can mean many things: simple awareness of an object, event or state, including awareness of a perception, thought, feeling, disposition, intention, action, or of one's habits of doing these things. It can also mean letting one's thoughts wander over something, taking something into consideration, or imagining alternatives. One can reflect on oneself reflecting.

Reflection does not necessarily imply making an assessment of what is being reflected upon, a distinction that differentiates it from critical reflection. Critical reflection may be either implicit, as when we mindlessly choose between good and evil because of our assimilated values, or explicit, as when we bring the process of choice into awareness to examine and assess the reasons for making a choice.

When the object of critical reflection is an assumption or presupposition (CP, A), a different order of abstraction is introduced, with major potential for effecting a change in one's established frame of reference. Assumptions upon which these habits of mind and related points of view are predicated may be epistemological, logical, ethical, psychological, ideological, social, cultural, economic, political, ecological, scientific, spiritual, or pertain to other aspects of experience. As adults, we can become critically reflective of our own assumptions as well as those of others. Critical self-reflection of an assumption (CSRA) involves critique of a premise upon which the learner has defined a problem (e.g., "a woman's place is in the home," so I must deny myself a career that I would love). Significant personal and social transformations may result from this kind of reflection.

Philosophical Foundations

Philosopher Harvey Siegal (1988) identifies the assessment of reasons as fundamental to the contentious concept of rationality. Reasons imply an assumption of some supporting principle. A reason is considered valid if it is consistent; the reason for an educator to make a judgment of a learner's progress is valid only providing the principles (criteria) used for doing so also pertain in judging the progress of other learners in the same situation. Critical reflection is principled thinking; ideally, it is impartial, consistent, and non-arbitrary.

There is an assumption of universal standards that constitute criteria allowing us to evaluate reasons. Principles pertaining to informal logic--making warranted generalizations, proper inductive or deductive inferences, avoiding fallacies, and so on--are general principles with universal relevance. Context-specific principles that govern the proper interpretation in physics, history, economics, art, literature, or music are of obvious importance to reason assessment in these fields.

Principles by which reasons are defined and assessed evolve as their defining traditions change. Siegal (1988) writes,

Principles embody rationality and define and assess reasons in a tradition at a time. As the tradition evolves, so do the principles which define and assess reasons. So what counts as a good reason in a tradition may change over time; today's compelling reason may be seen as less than compelling tomorrow Still the principles which determine the compellingness of

reasons at a time apply to all putative reasons impartially and universally; moreover, such principles still serve to embody rationality. (p. 136)

This is the way the universal constructs of reason assessment, CRA, and adult learning are interpreted differently by different cultures and in different time periods. Choice is rational if one chooses well. The culture and time determines what "well" means.

To enter into a discourse to assess alternative beliefs is to initiate a process that is predicated upon universal principles. Roochnik (1990) explains: Once the debate commences certain conditions implicitly become operative. Most important of these is that the opponents agree that in principle the debate can be won ... a standard must be evoked by which the two competing positions can be measured.., it makes victory--the necessary goal of such a game--possible. (p. 198)

In contemporary cultures, a person would be defined as rational who, on the basis of information available, assesses reasons logically and reflects critically in order to achieve the best foreseeable consequences of an action. Halstead (1994) argues that reason is dependent upon nonrational forces, including sentiment, instinct, conjecture, imagination, and experience.

Siegal recognizes that informal logic and critical reflection are essential dimensions of reason assessment. Most of us gain a working knowledge of informal logic and may acquire a disposition or skills for critical reflection through parents, culture, and education prior to arriving at adulthood. However, it seems to me that the activity of CRA and particularly its variant, becoming critically self-reflective of assumptions, are distinctly adult dimensions of critical reflection and of reason assessment. If we are to understand reason assessment as the sine qua non of rationality, reason assessment and rationality must be understood as involving informal logic and critical reflection, including CRA.

There are educators like Bowers (1977) and Hemphill (1994), and feminist educators like Michelson (1996), who worry that education for the enhancement of critical reflection and empowerment reflects an arbitrary cultural hegemony of an ideology of Western Enlightenment. However, Siegal argues that rationality and critical reflection do not constitute just another ideology, as these educators suggest. Ideology presupposes a commitment to rationality: the justification for embracing an ideology depends upon advancing and supporting reasons for doing so. Any analysis of ideology presupposes standards of rationality and a recognition of the cognitive force of reason. Siegal (1958) concludes:

We need not regard educational ideals as fundamentally ideological; nor must we regard critical thinking as itself ideologically bound or as tied to specific cultural or class interest.

Rather... critical thinking and rationality are basic in the sense that they alone allow for the possibility of the intelligent evaluation and criticism of ideology. To take the study of ideology seriously is to rely on the critical leverage afforded by critical thinking and (non-ideological) rational analysis. (p. 77)

"Postmodernist" writers like Derrida (1982), who considers rationality "logocentric," and Lyotard (1984), who derides all "metanarratives," are among those who challenge the universal validity of such concepts as rationality, reason assessment, and critical reflection. However, arguments against the universality of rationality and critical reflection themselves demonstrate the necessity of assessing reasons and becoming critical of assumptions. Once these critics choose to enter into rational discourse, they have no choice but to agree to observe universal principles of rationality.

Transformation Theory maintains that human learning is grounded in the nature of human communication; to understand the meaning of what is being communicated--especially when intentions, values, moral issues, and feelings are involved--requires critical reflection of assumptions. Should CRA and the verification of beliefs through discourse, both central to Transformation Theory, and indeed should rationality itself be understood as involving universal principles of human communication and learning? As we have seen, these principles transcend local culture, but a culture has the option to encourage or negate discourse.

Transformation Theory does not hold that critical reflection and rational discourse represent some transcendental version of the eternal verities or are of an order of reality that transcends the empirical world of change, but simply that they have been found to work better in more circumstances than have other options. They are universal in the sense that discourse is predicated upon universal principles. From this vantage point, critical reflection, discourse, and rationality itself develops only as a consequence of inquiry. The alternatives to discourse involve basing understanding upon tradition, authority, or physical force. In cultures where the objective is to perpetuate a religion or a regime, or to produce a docile workforce, critical reflection and discourse are commonly limited.

Similarly, the answer to the question raised by Transformation Theory of which frames of reference are more functional is those that work, that opinions they lead to are for the most part true or are more often true or justified than other opinions to which alternative ways of framing experience would lead. It is this logic that suggests that frames of reference should be considered more functional or more ideal when they are more inclusive, differentiating, critically reflective, open to other points of view, and integrative of experience. These are enabling conditions extrapolated from the universal principles of discourse.

If we are to fully comprehend when the meaning of what is being communicated to us includes feelings, values, ideals, moral decisions, and intentions, we must of necessity become critically reflective. We must not just critically reflect on the way we decode the meaning of words, but on the assumptions that the person communicating with us is being coherent (what is being communicated is appropriate to some set of norms), is telling the truth, is being truthful (not trying to deceive), and is being authentic in communicating feelings. Can you believe the person who tells you they love you? Is the person diagnosing your illness, building your house, giving a lecture, giving you orders, or reviewing a play qualified to do so? Is the friendly stranger really trying to proselytize or to sell you something? Is he or she making a sexual overture? Is the point of view expressed simply echoing some party line? We must ask, in the popular vernacular, "Where is this person coming from?" We cannot learn the meaning of what is being communicated without becoming critically reflective of sub-textual assumptions about truthfulness, truth, authenticity, and coherence.

The Development of Critical Reflection of Assumptions

Is there a developmental sequence in the evolution of CRA through the life span? King and Kitchener (1994) present considerable empirical evidence for the development of reflective judgment from childhood to adulthood. They define "epistemic cognition" as the process an individual invokes to monitor the epistemic nature of problems and the truth value of alternative solutions. As such, they are writing about CRA:

It includes a person's knowledge about the limits of knowing, the certainty of knowing and the criteria for knowing. It can be distinguished from cognition, the premonitored cognitive process on which knowledge of the world is built (such as reading, memorizing, and computing), and from metacognition, the cognitive processes invoked to monitor cognitive tasks. (p.12)

Learning about the limits, certainty, and criteria for knowing presumes CRA. If you say that it will rain tomorrow, I need to critically attend to assumptions regarding the source and authority for your opinion. Are you simply generalizing from your own past experience with the weather or are you reporting a forecast by a weather bureau that you have read or heard?

Research cited by King and Kitchener suggests that we may move through five sets of learning assumptions from childhood to adulthood. They present this as movement through developmental "stages" (see Figure 1), to reach, as adults, a sixth and seventh stage that represent "reflective judgment." One could contend that as we move toward more adult levels of understanding, each assumption constitutes a habit of mind.

It seems clear that movement through these ways of knowing, related to age and education, is toward a more inclusive, differentiating, permeable, critically reflective, and integrative frame of reference. Insofar as our circumstances permit, we naturally move toward frames of reference with these characteristics, to better understand our experience.

King and Kitchener's concept of reflective judgment implies a discursive rationality-arriving at agreement on meaning through discourse by giving and assessing reasons, critically assessing arguments and assumptions, examining evidence and seeking to validate beliefs and interpretations consensually. This view is consistent with Transformation Theory, which describes this as the process by which we validate what others mean when they communicate with us and justify the claims made by alternative beliefs.

King and Kitchener reserve the term "reflective" to describe the adult reasoning characteristic of Stages 6 and 7. It is at these levels that knowledge is understood not as a "given" but as actively constructed--knowledge claims are understood in relation to the context in which they were generated. While judgments must be grounded in relevant data, conclusions should remain open to reevaluation. "Context" can refer to historical or biographical factors pertaining to a belief or "knowledge claim," or to a taken-for-granted paradigm, system, or canon in which the belief is grounded.

From the view of Transformation Theory, this description of reflection as the active construction of knowledge claims, understood within the context of their origins, is a description of CRA. When a woman enters the process of discourse to justify what is for her the problematic belief that a women's place is in the home and with others critically examines the origins of this belief in their historical and biographical contexts, her understanding of the validity of this belief may be transformed.

Transformation Theory also recognizes the crucial role past emotional experience plays in learning--sending signals that facilitate our decision making by eliminating some options and highlighting others. Another important non-cognitive factor in transformative learning is the disposition and emotional stamina to believe that one has both the will and the way to reach his or her reflectively redefined goals.

Critical Reflection of Assumptions and Assimilative Learning

Learning may be understood as the process of using a prior interpretation to construe a new or revised interpretation of the meaning of one's experience in order to guide future action. The description of CRA as involving the active construction of knowledge helps us differentiate the learning dynamic that pertains in childhood from that which may become manifest through CRA in adulthood. When a child, in one of the early stages above, is made aware that his or her assumptions about knowing or behaving are no longer functional and a

more appealing model of interpretation or behavior is at hand, the movement toward the more functional behavior with its different set of assumptions will be unlikely to involve epistemic cognition at all. The learning process may involve only a relatively mindless response to social pressure or to please the teacher, parents, or one's peers. This response, however, does imply a kind of tacit judgment inasmuch as a decision is being made that an old way of thinking or behaving does not work in the specific situation, and a choice is being made to move to the way of thinking or behaving deemed more functional or acceptable.

This form of tacit judgment is not limited to learning in children. It characterizes a common way adults learn to adapt to change as well. Our situation changes, and, beyond our scope of awareness, we make a tacit judgment to move toward a way of thinking or behaving that we deem more appropriate to our new situation. This is the process of assimilative learning. Psychoanalysts use assimilation to mean adjustment or accommodation to a difficult situation by accepting it as conforming to one's desire.

Intuition is another tacit form of judgment that occurs without recourse to reasoning. Imagination, the capacity to seek out alternatives and to look at things as if they could be otherwise, is central to adult learning and to critical reflection. Imagination can involve either or both critical reflection or tacit judgment.

Taylor (1994) challenges the idea that transformative learning necessarily requires CRA. On the basis of a study of twelve Americans who had lived in another country for at least two years, Taylor contends that the development of new habits of thought and living in different cultures can produce a transformation in frame of reference without the person being aware of it. Taylor's analysis suggests that his subjects experienced dramatic changes in orientation based upon assimilation rather than critical reflection on assumptions. In a sense this assimilative learning, albeit mindless, did indeed result in major reorientation and changes in values. Dramatic changes in orientation may be effected by culture change as well as by brain washing, coercion, and indoctrination.

As I have used the term in the context of adult education, transformative learning refers to effecting transformations in frames of reference within the scope of one's awareness through CRA. It is entirely possible that a progressive sequence of related tacit judgments, acquired through assimilation, might lead to CSRA and a mindful transformation in frames of reference. For adult education researchers concerned with identifying transformative learning, the distinction between assimilation and CRA is a crucial one.

I would argue that CRA and its variant, critical self reflection on assumptions, are the emancipatory dimensions of adult learning, the function of thought and language that frees the learner from frames of reference, paradigms, or cultural canons(frames of reference held

in common) that limit or distort communication and understanding. If learning to think for oneself--freeing oneself from our conditioned assumptions about the world, others and ourselves--is essential in the world of work, in functioning as a citizen in a democracy, and in making responsible moral decisions in fast changing societies, then adult educators had better understand the central role played by CRA. It seems important for educators to note that there are several different uses of CRA common in adult learning with different implications for educational intervention.

Critical Reflection of Assumptions: A Taxonomy

It is important to make a distinction between the use of CRA to analyze a text or problem pertaining to improving performance, involving instrumental learning-"objective reframing"--and CSRA that involves "subjective reframing."

Objective Reframing

Narrative CRA involves critically examining the validity of the concepts, beliefs, feelings or actions being communicated to you (in speech, books, paintings) by assessing the truth or justification of taken-for-granted assumptions. Narrative CRA is a common classroom goal in graduate adult education. We critically examine the assumptions of established definitions, theories, and practices to better understanding the paradigms and canon that their writings represent. Narrative CRA is the stuff of literary criticism. Art criticism includes CRA pertaining to the standards of aesthetic judgment.

Action CRA involves a pause in task-oriented problem-solving to critically examine one's own assumptions in defining the problem, sometimes via a "generative metaphor" (Schon, 1979), in order to take more effective action to solve it. One can be critically reflective of the content of the problem, the process of problem-solving or the premise of the problem. Becoming critically reflective of the premise or assumption and defining the problem can lead to a transformative redefinition of the problem, or to "problem posing." Sometimes this involves drawing upon a more experienced person's broader repertory of ways to conceptualize problem posing in order to redefine an operational problem. The objective is instrumental: to improve performance.

Revins' (1982) "action learning" brings persons with different corporate specializations or from different organizations together to study a particular corporation's problem in order to introduce new perspectives that may lead to new ways of posing and solving the problem. The emphasis is on the introduction of new points of view and habits of mind in order to solve a problem rather than on engaging in a critical analysis of the reasons one has used to frame the problem in the first place. Action CRA pertains to instrumental learning. Schon (1983) offers a perceptive analysis of how Action CRA works.

Subjective Refraining

Critical self-reflection on assumptions (CSRA) emphasizes critical analysis of the psychological or cultural assumptions that are the specific reasons for one's conceptual and psychological limitations, the constitutive processes or conditions of formation of one's experience and beliefs. CSRA involves what Steedman (1991) calls a "stereoscopic" account of the construction of meaning: "one lens is on the individual context of interpretations, the other is on the social context which creates the individual" (p. 55). CSRA is the process involved in psychotherapy, conscientization, consciousness raising, and other approaches to effecting major perspective transformations. I have elsewhere reported on studies that identify common phases in this learning process (Mezirow, 1991). There are four common forms of CSRA familiar to adult educators.

Narrative CSRA is the application of Narrative CRA to oneself. Maxine Greene (1990, p. 254) has described this as an "efferent" transaction, a carrying-over of insight gained from a narrative into the learner's lived experience. Such critically self-reflective insight can lead to an analysis of one's own unwarranted assumptions pertaining to the subject content of the narrative. Narrative CSRA may result in other forms of CSRA; in practice, there may be considerable overlap among these forms.

Systemic CSRA involves critical reflection on one's own assumptions pertaining to the economic, ecological, educational, linguistic, political, religious, bureaucratic, or other taken-for-granted cultural systems. We critically reflect on the canons, paradigms, or ideologies that have generated traditional roles and relationships, and on how they have shaped and limited the development of our point of view and have fostered dependency relationships. Systemic CSRA often leads to some form of collective or collaborative social action. This is the focus of Freire's (1970) educational ideas, consciousness raising in the women's movement (Hart, 1990), and CSRA in other social movements.

Organizational CSRA is primarily directed at identifying assumptions that are embedded in the history and culture of a workplace, and how they have impacted on one's own thought and action. Argyris and colleagues (1985) define "doubleloop learning" as learning that is "concerned not with choosing among competing chains of means-ends reasoning with a given set of standards, but choosing among competing sets of standards ('frames' or 'paradigms')" (p. 86). This is critically reflective learning.

Argyris and Schon (1974) have identified the most common frames of reference or "theories-in-use" that inhibit such learning as: (a) achieve the objective as the actor defines it, (b) win, do not lose, (c) suppress negative feelings, and (d) emphasize rationality (use cool reason to persuade others). These values lead to instrumental learning and strategies of seeing

interpersonal interactions as winlose games and secret efforts to manipulate others. The result of this perspective is to limit learning to strategies and tactics for achieving one's own objectives.

The organizational norms that commonly inhibit critically reflective learning include: let buried failures lie; keep your view of sensitive issues private; enforce the taboo against public discussion; do not surface and test difference concerning organizational problems; avoid seeing the whole picture so one does not see how problems are connected; protect yourself by avoiding interpersonal confrontation and public discussion of sensitive issues; protect others in the same way; control the situation and the task by making up your own mind and keeping it private; and avoid public dialogue that might refute your view. If transformative learning is important, then traditional mind sets of this kind must be identified and replaced with a more open model of communication. The evolving concept of the learning organization embodies CRA, but seldom CSRA. (Marsick, Cederholm, Turner, & Pearson, 1992; Watkins & Marsick, 1993).

Moral-ethical CSRA involves a critique of the norms governing one's ethical decision-making. This is critical reflection on value judgements one has made or is considering, and is often related to conscience and one's idealized self-image. In Transformation Theory, challenges to the validity of a particular norm are met through critical discourse, a collective process of assessing reasons, examining assumptions, weighing evidence, hearing arguments, and arriving at a tentative best judgement until other analyses, arguments, or perspectives are encountered. The process of establishing the justification for a norm is open-ended.

Therapeutic CSRA creates awareness and critical insight pertaining to assumptions governing one's problematic feelings and related dispositions, and their action consequences. It is a type of problem-posing and problem-solving in which one examines the sources, nature, and effect of assumptions governing the way one feels and is disposed to act upon his or her feelings. This generic learning process is central to psychotherapy, and its facilitation has generated sophisticated theoretical explanations, research, and modes of professional intervention in that field.

Cognitive psychotherapy (e.g., Kurhlwein & Rosen, 1993) is the orientation of psychologists closest to educational interventions in other uses of CSRA. Among the major differences between adult education and therapy is that adult education is concerned with a much wider range of learning content and applications of CSRA than is therapy. Therapy has historically been located in a medical context, therapists deliberately use transference to foster learning, and the doctor-patient expectations of many learners in a therapy situation probably make it more difficult to establish a subject-subject relationship than in an adult education context.

Gould (1978, 1990) has elaborated upon the nature of the most commonly encountered psychological problems of adulthood requiring Therapeutic CSRA and has delineated elements of a short-term "therapeutic learning program" (1990, pp. 134-156). It is important to make some distinctions here between psychologically-oriented programs that foster self-reflective or critically self-reflective learning and those involving Therapeutic CSRA. We may become self-reflective of the way we tacitly react to an authority figure by participating in a "T-group" or of our aptitude or learning preferences as we come to see these reflected in our performance on the Myers-Briggs test or on some other psychological instrument that places the learner in predefined prototypical categories, a practice many adult educators consider undesirable.

We may also become critically self-reflective by becoming aware that we are making a premature value judgment or are being inconsistent in acting out our values. This critical self-reflection involves the kind of learning associated with some forms of sensitivity training and a legion of psychologically-oriented short term workshops or laboratories. Sometimes participants are aided to see their old habits of reaction in light of generalized research findings or a maxim to live by, like "take responsibility for yourself. Such learning may result in potentially valuable, though limited, insight into one's established patterns of reaction. They do not necessarily involve critical self-reflection of one's own assumptions.

Epistemic CSRA. The learner sets out to examine the assumptions and explore the causes (biographical, historical, cultural), the nature (including moral and ethical dimensions), and consequences (individual and interpersonal) of his or her frames of reference to ascertain why he or she is predisposed to learn in a certain way or to appropriate particular goals. This involves making explicit one's process of epistemic cognition. The reframing process is not directed at solving an immediate problem but on explicitly identifying one's frames of reference, their sources, and implications for understanding the world, others and oneself as an educator, manager, planner, or in other roles. This may involve a special kind of life history inquiry (Dominice, 1990). Issacs (1993) refers to Epistemic CSRA as "triple loop learning."

The Dialogue Project at MIT's Center for Organizational Learning is a program designed to facilitate Epistemic CSRA. The program draws upon the work of physicist David Bohm to suggest that, through discourse or collaborative, critically reflective dialogue, it is possible to kindle a new mode of paying attention, to perceive--as they arise--the assumptions taken for granted, the flow of the polarization of opinions, the rules for acceptable and unacceptable conversation, and the methods for managing differences. And since reflection, by its nature, looks back at what has already taken place, it is innately limited for anticipating assumptions, opinions, rules and differences that are only now emerging. The mindfulness embodied in dialogue involves awareness of the living experience of thinking, not reflection after the fact

about it. For us to gain insight into the nature of tacit thought, we must somehow learn to watch or experience it in action. (Issacs, 1993, p. 31)

The Dialogue program brings learners together in a protected environment without agenda, leader, or task to be performed. A facilitator initially helps the group focus on collective thought and share assumptions that are major factors in making their judgments. Topics emerge as participants learn to understand how others think and feel about common concerns and about each other. The emphasis is on helping learners understand the ways their own habits of expectation and patterns of thought and interaction deeply influence their experience. Issacs (1993) writes, "dialogue involves learning about the context and nature of the processes by which people form their paradigms and take action" (p. 38).

For this kind of educational experience to achieve its promise, it must go far beyond the insights normally associated with a traditional group dynamics learning experience; the content of the discourse must be a focused, critically self-reflective, collaborative inquiry into how one's own habits of mind have framed his or her points of view. Little is known about such programs and there is a great need for research on this model of adult education.

The AEGIS program at Teachers College, Columbia University, is a doctoral program for adult educators that was designed to foster learning in each of the forms of CRA described above. In AEGIS, learners participate in courses with different emphases on fostering objective or subjective refraining throughout the curriculum (Mezirow, 1991). Educational approaches for facilitating and precipitating critical reflection (Brookfield, 1987,1991,1995; Cranton, 1994, 1995; Mezirow & Associates, 1990) provide additional discussion of discourse and reflective action.

The taxonomy presented above is derived from a review of different foci of CRA. Taking a somewhat different tack, Brookfield (1995) identifies three types of assumptions for critical reflection: paradigmatic assumptions that structure the world into fundamental categories (the most difficult to identify in oneself), prescriptive assumptions about what we think ought to be happening in a specific situation, and causal assumptions about how the world works and how it may be changed (the easiest to identify).

Critical Reflection of Assumptions and Discourse

Critical reflection may occur outside or within a discursive group. Discourse is understood as that special function of dialogue devoted to presenting and assessing the validity of reasons by critically examining the widest possible range of evidence and arguments in the context of attempting to find understanding and agreement on the justification of beliefs. CRA may be a product of discourse, but it may also be the way an individual learner gains insight in dealing with a disorienting dilemma.

A learner may acquire a potentially transformative learning experience through reading or viewing art. To be able to act with confidence on the insight, he or she must submit it to either a discursive assessment of whether the insight is justified or, in the case of instrumental learning (controlling or manipulating the environment), submit it to an empirical test to determine whether it is as the learner purports it to be. You may have an especially brilliant insight beyond the understanding of those participating in the discourse, but this only means that you must seek validation through further discourse with a broader or more informed group of participants or a through discursive assessment at a later point in time. The only other ways you can justify your belief is by appealing to an authority, tradition, or the use of force to secure agreement.

The cultural canons governing the discourse in a particular subject matter field--and reflecting implicit boundaries and goals for discourse--must also be subjected to CRA. Such assumptions on boundaries and goals must always be open to challenge and to reason assessment so that a learner may avoid the tunnel vision of a particular canon. The justification for a valuable reflective insight through discourse may depend upon challenging a widely accepted canon. Recognizing this and assessing its implications can be crucial in moving the discourse ahead. A best judgment of trusting, informed, relatively objective and rational persons is always contingent upon further inputs of evidence or argument by a larger and more diverse group of participants. Validating a belief through the widest possible agreement is a developmental process, not a one-shot effort at securing consensus.

Conclusion

Learning to think for oneself involves becoming critically reflective of assumptions and participating in discourse to validate beliefs, intentions, values and feelings. Adult educators who are concerned with facilitating meaningful adult learning need to understand the significance of CRA and variations in how CRA is used for different purposes. This article attempts to clarify philosophical, developmental, and educational issues raised by this concept. The taxonomy presented differentiates between applications of CRA found in the literature of adult education, and suggests educational approaches for facilitating and precipitating this kind of learning. The professional task ahead is to find ways to translate the concept of CRA and discourse into curricula or programs, instructional methods, materials development, and evaluation criteria.

Figure 1: Stages of Development According to
King and Kitchener (1994, p. 208)

Stage 1 Single concrete category of knowing. Certain knowledge

gained by direct personal observation and needs no justification.

Stage 2 Two concrete categories of knowledge. A person can know with certainty through direct observation or indirectly through an authority.

Stage 3 Several concrete categories of knowledge are interrelated. Knowledge is assumed to be either absolutely certain or temporarily uncertain. Justification is based on authorities' views or what 'feels right.'

Stage 4 Knowledge is understood as a single abstraction. Knowledge is certain and knowledge claims are assumed to be idiosyncratic to the individual.

Stage 5 Two or more abstract concepts of knowledge can be related. Knowledge is seen as contextual and subjective. Beliefs are justified by using the rules of inquiry for the appropriate contexts.

Stage 6 Abstract concepts of knowledge can be related. Knowledge is actively constructed by comparing evidence and opinion on different sides of a issue; solutions are evaluated by personally endorsed criteria.

Stage 7 Abstract concepts of knowledge are understood as a system. The general principle is that knowledge is the outcome of the process of reasonable inquiry for constructing a well-informed understanding.

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